

Unit Design II: Piping and Instrumentation Diagram Report of Nitration Reactor

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1 Executive Summary

In the unit design I, the discussion and details are about reaction kinetics, reactor sizing, and so on. However, chemical engineering industry needs to put the numbers back to reality to specify the process with considerations of control system and safety system. To do that, the first thing is determining the relevant main disturbances which are the variation of feed flow compositions and the ambient temperature. After that, the variables that control loops need to identify the sensitive variables like reactor temperature and liquid level with identifying other variables like manipulated variables and measurement variables by using feedback and feedforward control strategies respectively in **Table 2.2**. More than that, there are additional features what been discussed include the locations of measurements and the failure position of control valves. However, the most important thing is identifying the types of controllers since they are deciding the stability and efficiency of control system by individually discussing pros and cons of Proportional, Integral, Derive, and Proportional-Integral controllers. After the basic details of control system, the interactions and stability issues between reactor temperature loop and liquid level control loop are evaluated by discussing output flowrate's controlling details. In the end of control system analysis, the solutions of main disturbances are evaluated by using feedforward control strategy to smooth the variation in feed composition before it influences results.

After comprehensively identifying the details of control system, the Pipe and Instrument Diagram of this nitration reactor **Figure 3.1** is drawn with three main elements like the Continuous-Stirred-Tank reactor R100, the condenser C100, and the output pump P100 with establishing reactor temperature control loop and level control loop by using two analyzers and various transmitters to comprehensively make decisions of flowrate of coolant fluid and output flowrate in a balance of reactor temperature, inlet and outlet composition, cooler temperature, and inlet flowrate. Then, a detailed safety instrumentation system is adding to the P&ID with considerations of emergency shutdown, alarms, relieve valves, and maintenances. More than that, an additional implementation of pressure Safety Instrumented Function is added to the P&ID and be drawn as the red part. In the end, the second sheet of P&ID shows specific symbols of valves, pump, reactor, strainer, and lines, line numbering method with fluid name and abbreviations, and a table of instrument identifiers is built up in **Figure 3.2**.

In the Safety Integrity Level analysis part, the main hazards are identified to be high reactor temperature, high pressure, and low coolant flowrate what all lead to low efficiency and dangers like explosion. Before using fault tree analysis to find the system's safety integrity level (SIL), the target of SIL is established by using Risk graph with using four parameters, then, by comparing the fault tree analysis result with the risk graph result, the control system cannot meet the target. In this case, an additional pressure alarm system is added as the additional safety instrumentation function in the Figure 3.2 P&ID graph to decrease the probability of failure on demand to meet target SIL. Then, as the proof, the protected fault tree is present at part 4.5 what indicate the target SIL is meet. Additionally, to find potential problems, a specific justification of failure rate's reliability and limitation is been discussed.

In the end, this unit design as the part of the plant design can bring local society economic opportunities include more working opportunities, tax, local supply chain, and so on. However, it also bring environmental and safety concerns to the local community, so regulatory compliance, risk assessment, hazard mitigation measures, and community engagement needs to be taken to avoid.

Table of Content

1 EXECUTIVE SUMMARY	2
2 CONTROL SCHEME	4
2.1 MAIN PROCESS DISTURBANCES.....	4
2.2 CONTROL LOOPS	4
2.3 FEATURES OF CONTROL LOOPS	5
2.4 CONTROLLER TYPES.....	5
2.5 CONTROL SYSTEM INTERACTIONS AND STABILITY ISSUES	6
2.6 SOLUTIONS OF DISTURBANCE	6
2.7 ALARM SYSTEM.....	6
3 PIPING AND INSTRUMENTATION DIAGRAM.....	7
3.1 P&ID DIAGRAM	7
3.2 P&ID LEGEND	8
4 SAFETY INTEGRITY LEVEL ANALYSIS.....	9
4.1 IDENTIFICATION OF HAZARDS	9
4.2 QUALITATIVE PARAMETERS OF THE RISK GRAPH	9
4.3 UNPROTECTED FAULT TREE ANALYSIS.....	10
.....	10
4.4 IMPLEMENTATION OF SAFETY INSTRUMENTED FUNCTION.....	11
4.5 PROTECTED FAULT TREE ANALYSIS	11
.....	11
4.6 JUSTIFICATION OF THE USED FAILURE RATES.....	12
5 SUSTAINABLE REFLECTION.....	13
6 CONCLUSION	13
REFERENCES	14
APPENDIX	15

2 Control Scheme

2.1 Main process disturbances

During making control system around this reactor, the control objective is targeted to be ideal conversion rate. Based on that target, there are two formulations can be used to find what variables can influence conversion rate which are energy-balance conversion and mole-balance conversion. For energy-balance conversion, its primary variables are cooler's temperature, feed flowrate, compositions of feed flow, and reactor temperature. For mole-balance conversion, its primary variables are feed flowrate, compositions of feed flow, and reactor temperature. The most variables like cooler's temperature, feed flowrate, and reactor temperature where the compositions of feed flow are not controllable but it indeed influence whole system's product, in this case, the variation of feed flow compositions will be the main process disturbance. However, even there is an absence of ambient temperature in formulations of conversion rate, it indeed is important to conversion rate. High ambient temperature will increase energy consumption of cooler and making ideal conversion rate easier to reach at the beginning, in this case, the ambient temperature as the external condition cannot be controlled naturally and will bring some effects to the control system so it is another main disturbance variable.[1] In the end, after analyzing formulations of control objective, the conversion rate, and engineering considerations, the variation of feed flow compositions and the ambient temperature are the two main process disturbances in this unit's control system.

2.2 Control loops

After defining basic control variables, there are two main controlled variables to design two control loops, conversion control loop and liquid level loop.

The first control loop is conversion control loop, in this loop, the conversion is controlled by the temperature of cooler and output flowrate what are the controlled variables and the feed flowrate what is the manipulated variable by using the measurement of temperature of reactor, variation in feed composition, ambient temperature, and conversion of reaction what are the measured variables. Due to the measured variables, variation in feed composition and ambient temperature, are disturbance variables, Feedforward control strategy can be selected to offer faster response times by addressing disturbances before they affect the system and reduce the burden on feedback control by mitigating disturbances proactively by using information about disturbances to adjust control variables preemptively. In this case, this conversion control loop has all disturbance variables what needs fast response strategy to adjust control variables preemptively, so Feedforward is selected to be used in this loop.

The second control loop is liquid level control loop, the level control is involved with the formulation of macroscopic mass balance, so the controlled variable is the output flowrate by using measurement from level of liquid and set point data from manipulated variable, the feed flowrate. The level control is aiming to stay vessel's liquid level to be safe that no dangers like explosion or overloading will happen, so it is important to consider but it does not require fast response strategy so the basic scheme Feedback can be used. Feedback is the most common used control strategy in process control designing bases on its simplicity of controlling dynamic-behavior system by comparing the actual measurement data and the desired data to output signal to adjust differences.

Based on above analysis, the Table 2.2 of each control loops' controlled variables, measured variables, and manipulated variables is generated. Meanwhile, the conversion control loop is using Feedforward control strategy due to the disturbances and the liquid level control loop is using Feedback control strategy due to less complexity, so the overall control scheme of this reactor is a combination of Feedforward and Feedback.[2]

Name of Control Loop	Controlled variables	Manipulated variables	Measured variables	Strategy
Conversion	Temperature of cooler; Output flowrate	Feed flowrate	Temperature of reactor; ; Variation in feed composition; Ambient Temperature; Conversion	Feedforward
Level of reactor's liquid	Output flowrate	Feed flowrate;	Level of reactor's liquid	Feedback

Table 2.2: Conversion control loop's and liquid level control loop's controlled variables, measured variables, manipulated variables, and control strategies

2.3 Features of control loops

After identifying this unit's basic control loops with their corresponding measured variables, controlled variables, and manipulated variables, there are additional information and details need to be specified like measurement location, measurement way, and failure position of control valves.

For conversion control loop, there are needs of measuring temperature of reactor, variation of feed compositions, ambient temperature, and conversion rate. For the locations of measurements of reactor temperature and conversion, their locations should be outflow because the output flow defines almost overall, more steady, accurate, and final data of reactor temperature and reaction conversion. For the location of measurement of variation of feed composition, it needs to be located at the beginning of process which means before the feed flow control valve so that the control system can adjust the disturbance before it makes the disturbance to be unchangeable, while the ambient temperature is supposed to be away of this reactor so that the heat effect cannot influence the measurement. Meanwhile, the way of measurement also defines the accuracy and precision of data. For both two temperature variables, namely reactor temperature and ambient temperature, they can be detected by using temperature sensors, however, for variation of feed composition, it needs to use inline analyzer to find compositions by using spectroscopy to monitor the composition of real-time flow, and to ensure its accuracy since it defines whole reaction's effectiveness, another measurement way is finding the fluid density and do thermal conductivity measurement bases on known main chemicals to calculate the composition of real-time flow. For conversion rate, inline analyzer also be implemented but another way is different what is finding the heat of reaction to find corresponding conversion. [3]

For liquid level control loop, the only variable needs to be measured is reactor's liquid level. By considering the type of this reactor is Continuous-Stirred-Tank reactor, the location of measurement should be above blenders and has less sprays, in this case, the measurement can be considered to be the top of reactor tank. By using a radar level transmitter to emit microwave or radiofrequency pulses that propagate through the vapor space above the liquid in the reactor, the liquid level can be estimated. Additionally, the failure positions of this unit pipeline's control valves include feed flow, output flow, and coolant fluid flow control valves also need to be determined. For feed flow control, the failure position is close when the signal or power is lost to avoid overloading situation to have explosion or doing emergency drain. For the outflow control valve, it will be open if signal is lost or unordinary to do emergency drain as fast as it can. For the coolant control valve, it will stay in its last position so that the reactor will not be over heated.[4]

2.4 Controller types

Now, after defining additional features of control loops, the type of controllers in the control loop needs to be specified. In chemical industry, there are few types of controllers include proportional (P) controller, integral (I) controller, derivative (D) controller, a combination of both three controllers proportional-integral-derivative (PID) controller or proportional-integral (PI) controller, and on-off controller to control the system bases on different needs. For this control system, there are conversion control loop and level control loop what has cooler temperature & output flowrate and output flowrate respectively, where the output flowrate variable is involved with two control loops will require more comprehensive controller. To explain more, the situation would be that decisions of potential output flowrate controller will process two functions at same time which means oscillations conducted from conversion function and mass balance function, so the controller needs to create less oscillations which means P, I, or D controllers cannot be used for this occasion. This bring PI and PID controllers to be potential candidates, but due to PID controller has an additional function of derivative by comparing with PI controller, PID controller can lead to instability or oscillations, particularly in systems with aggressive tuning or under certain operating conditions, however, the instability in industry bring it to be high potential cost which indicates PI controller is more acceptable for output flowrate due to its fast action on correction and eliminations on steady-state errors and its simplicity. For controller of cooler temperature, it needs to process the measurements from reactor temperature, variation in feed composition, ambient temperature, and conversion rate what indicate **high tense** of noises, so that PI controller still be the appropriate candidate for cooler temperature controller to eliminate measurement errors and noise and fast response to do correction action at the same time with more simplicity and high stability. [5]

2.5 Control system interactions and stability issues

After fulfilling details and features of control loops, the possible interactions and safety issues between these conversion control loop and liquid level control loop. During controlling this unit system to pursue high conversion possibility base on feed conditions, conversion and liquid level loop are both controlled by output flowrate to control reaction's resistance time and accumulation what lead to different requirements of output flowrate. Here is an example to show the interaction, in a scenario, the liquid level is nearly high to the set limit but the resistance time is too high that the feed flowrate is increasing or output flowrate is decreasing, so that the liquid level will reach the limit and the emergency trigger is supposed to be activated so outflow flowrate is increasing to its maximum, but due to the resistance time requirement, the control system may decide to keep increasing the feed flowrate and decreasing the out flowrate where the overloading or even worse the explosion will happen. In order to keep control system to be sensible running, the direct and indirect interactions should be identified. The above is one of this system's indirect interactions what occurs when changes in one control loop affect the process dynamics, which in turn influence other control loops, however, this system's direct interaction can the change in manipulated feed flowrate will influence both loop's decision so the liquid level controller may increase output flowrate and the conversion controller decides to decrease the cooler temperature. Both of these two kinds of interaction made system stability to be an issue where the direct interaction will lead to indirect interaction what may make the process to be emergency shutdown or cause dangers of productions. For example, the conversion control loop decides to decrease the output flowrate to increase resistance time what lead to the liquid level to be increased and the cooler temperature to be less, then, the liquid control loop decides to drain the tank to avoid potential explosion so that the reactor is closed what made whole plant to be closed and high economical lost.

To solve the stability issue, setting sensible stability margin is a solution for it. The following steps can be taken like using frequency response techniques to analyze the system's stability characteristics, determining the desired gain margin in decibels and phase margin in degrees based on the system's requirements, adjusting controller parameters iteratively to achieve the desired stability margins while maintaining satisfactory control performance, simulating the system's response to various inputs and disturbances to validate the achieved stability margins and ensure robustness, and documenting the achieved stability margins and controller settings for future reference. [6]

2.6 Solutions of disturbance

As mentioned before, to solve disturbance issue, the feedforward strategy is used at conversion control loop where has two main disturbances of this reactor include variation in feed compositions and ambient temperature. By using the feedforward strategy, the controller will the measurements of disturbance variables feed composition and ambient temperature to incorporate their effects into the control algorithm to generate a preemptive corrective action. For example, the feed composition and the ambient temperature are lower than expected data, so the transmitter detects the change and sent it to the cooler temperature controller and the output flowrate controller to increase reactor temperature and make resistance time to be longer, then, then, before the variation in feed composition changed the conversion, the control system will preset all controller to overcome the disturbance. [7]

2.7 Alarm system

Alarm system is a system consist with level of sequence of alarms for safety assurance, process optimization, risk reduction, compliance with regulations, displaying design, and alarm rationalization. In chemical industry, installing alarms of dangerous variables to help workers to detect deviations from normal operating conditions promptly and to ensure the safety of personnel, equipment, and the environment by displaying them in the human-machine interface (HMI) in control rooms and operator consoles, in this case, workers and managers can take steps to process optimization by identifying inefficiencies, deviations, or equipment malfunctions and mitigate risks associated with process upsets, equipment failures, or human errors, more importantly, it can help company stakeholders to meet legal requirements by providing mechanisms for timely detection and response to deviations from regulatory thresholds or safety limits. In this reactor, there two variables need to be installed with alarms which are reactor temperature and liquid level to avoid phase change due to high reactor temperature to have possibility of explosion. [8]

3 Piping and Instrumentation Diagram

3.1 P&ID diagram [9]

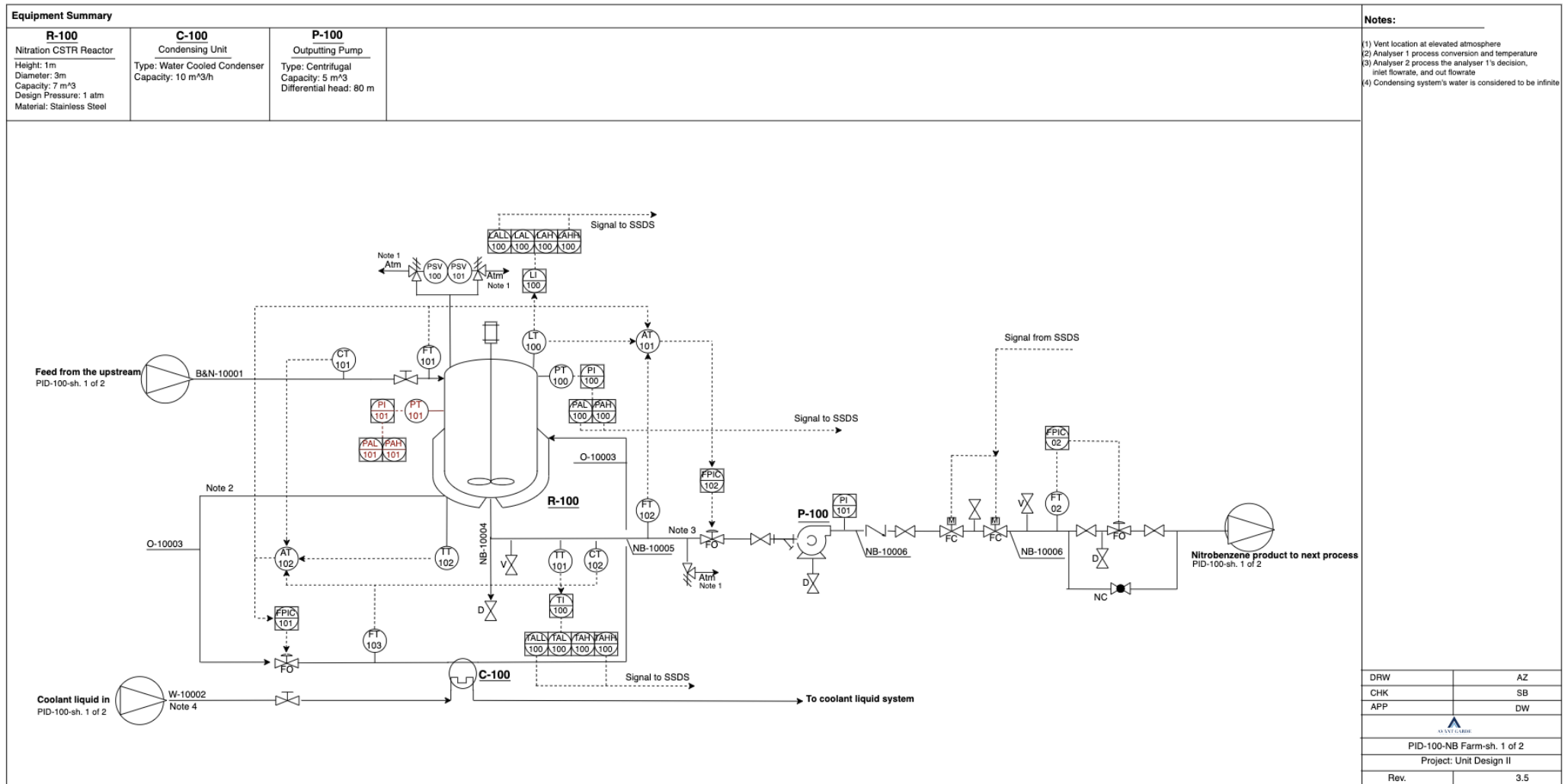


Figure 3.1: Piping and Instrumentation Diagram

3.2 P&ID legend

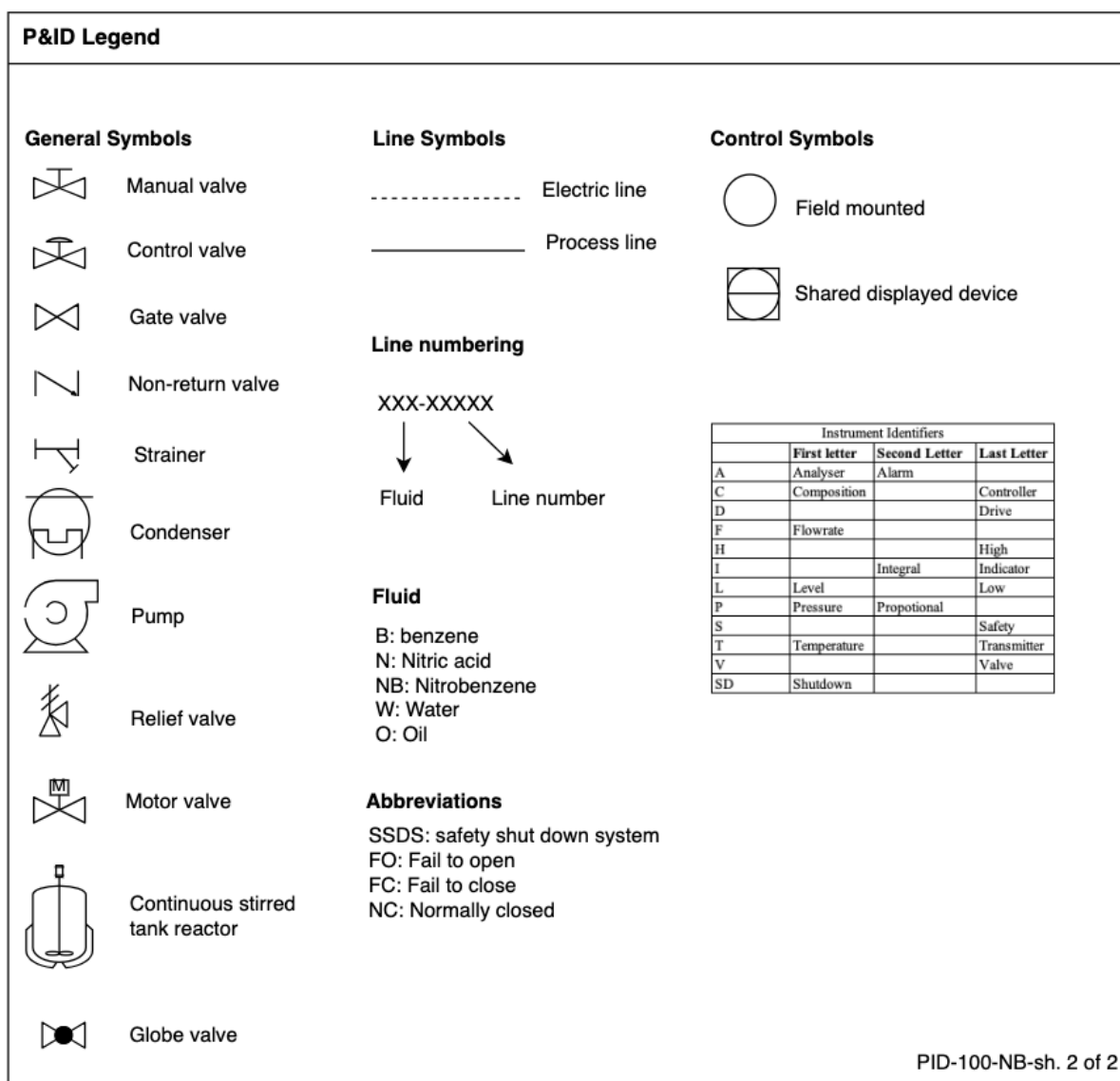


Figure 3.2: P&ID Legends

4 Safety Integrity Level Analysis

4.1 Identification of hazards

With this reactor unit, there are 3 main hazards include high reactor temperature, low cooler flowrate, and high reactor liquid level by evaluating their respective severity and likelihood. Now, each of them needs to be specified along to rank their significant level of this system for further Safety Integrity Analysis (SIL).

For high reactor temperature, due to the type of this nitration reaction is an exothermic reaction, the process is keeping releasing chemical energy to thermal energy which requires an additional cooler system to condense the temperature. However, the high temperature can undermine the efficiency of reaction by taking undesired reaction kinetics, more importantly, the high temperature will lead the inside chemicals to meet the phase change point so the supposed liquid phase will transfer to gas phase what will make the reactor pressure high, to some extent, the reactor will be exploded and bring high dangers to workers and community. In this case, since the process is in high demand, the likelihood of unexpected high reactor pressure will be medium to be Level C, and the category of it is 'Very High' by considering its severity is Level 5. Meanwhile, the above hazard also brings the condensing system considerations to the front. The condensing system helps to control the temperature of reactor not by controlling the condensing coolant fluid's temperature but by controlling the flowrate of coolant fluid. It seems that this hazard has the equivalent hazard category, however, the low condensing flowrate as the hazard's severity not highly and directly high to be considered as Level 3, but its likelihood level is same as the high reactor temperature's one to be Level C, in this case, the hazard category of low condensing coolant flowrate is 'Medium'. Now, there is another way that dangers will be created by overloading the reactor. During the process, there is a macroscopic mass balance is taking the main role which involves with the feed flow and the output flow, in this case, the level of reactor liquid is the representation of accumulation situation. There are two accumulation situations need to pay attention includes high liquid level and low liquid level. For high liquid level, it means the reactor is reaching overloading situation that the reactor can have high liquid-pressure what can lead to reactor explosion, so its severity is Level 5 with low likelihood Level B, so its hazard category is 'High'. For low liquid level, the low efficiency is brought up because the resistance time and product amount will not be desired, but its severity and likelihood are both not high, so it can be considered as not important enough.

Lastly, high reactor temperature hazard is considered as 'Very High' category, where high liquid level is ranked after it to be 'High', then, low condensing coolant fluid flowrate take the 'Medium' hazard category to be the last one. In the following parts, the high reactor temperature hazard will be significantly discussed to find its solution.

4.2 Qualitative parameters of the Risk graph

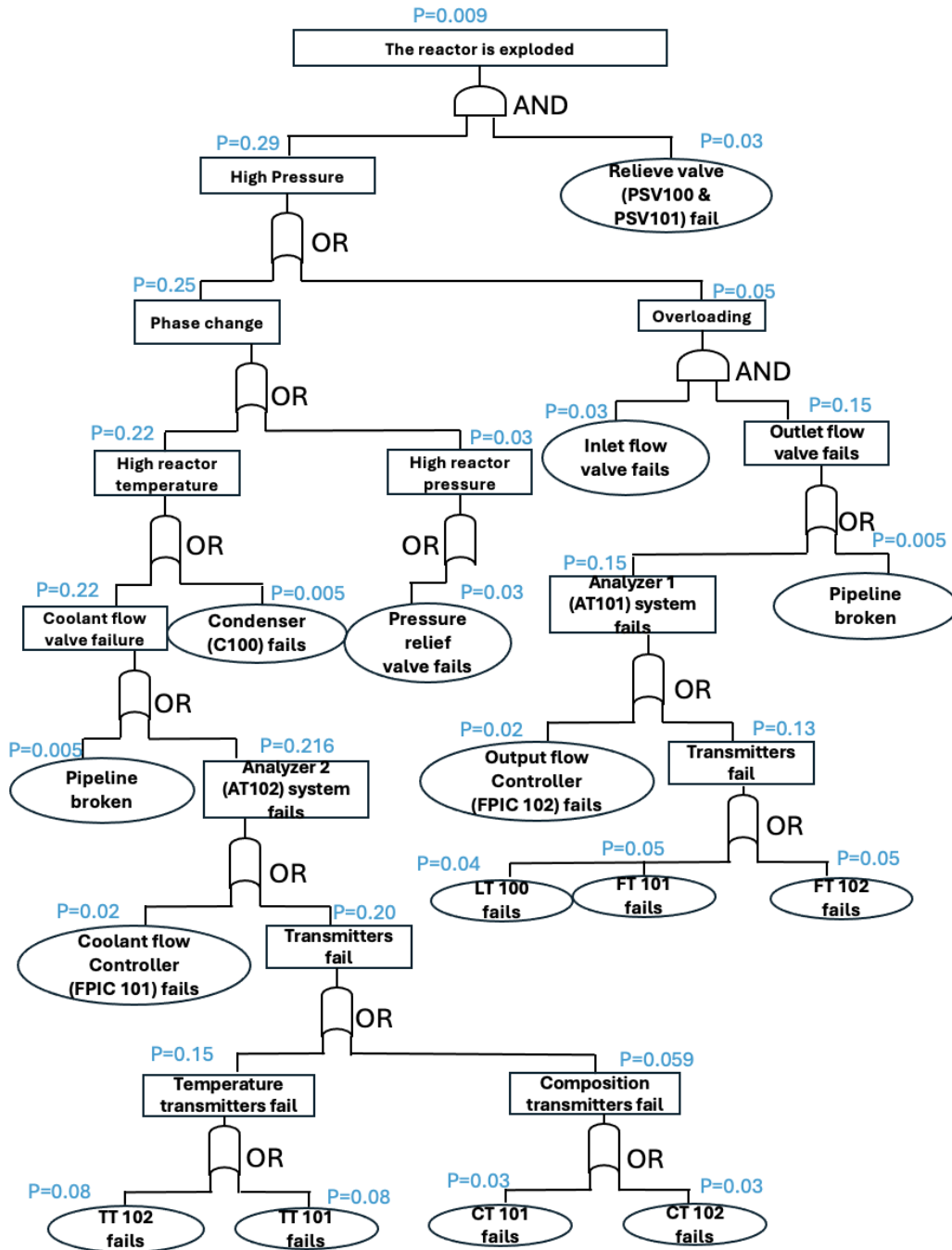
In theory, a process's safety system needs to be comprehensive enough to meet the highest safety requirement to avoid as possible as the designer can do. However, in the most situations of real engineering industry, the designer cannot use 30% more cost to meet the 0.1% more safety extent, so the extent of safety integrity level to the appropriate requirement of system's safety needs to be evaluated of the above analyzed 3 hazards by using four parameters include Consequence C_i , Frequency F_i , Probability of avoidance P_i , and Demand rate W_i .

For the high-reactor-temperature hazard, the too high reactor temperature always led to serious dangers like explosion so the consequence can be C_D which means the probability of loss life to be over 1, and the exposure of explosion highly-probably covers the operators so the frequency should be F_B , and the reactor temperature control loop can stay the probability of avoiding hazard to be higher but not totally reach the all conditions of being P_A , so it is P_B . Then, the demand rate is expected to be low by considering the control loops of level and reactor both take care of the hazard, but it can be evaluated later by specifying the event's PFD in the below unprotected Fault Tree Analysis.

For the low-coolant-flowrate hazard and the high-liquid-level hazard, they are kind of similar with the high-reactor-temperature hazard, but they are equipped comprehensive alarms and control functions, so in the end, the low-coolant-flowrate hazard's SIL is 1 (C_C, F_B, P_A, W_1) and the high-liquid-level hazard's SIL is 1 (C_C, F_B, P_A, W_1). In this case, the high-reactor-temperature hazard needs additional implementation of safety instrumentation function after doing fault tree analysis by evaluating the possible solutions.

4.3 Unprotected Fault Tree Analysis

To build up the Fault tree to do further analysis to find the safety integrity level, the top event is specified as the high-reactor-temperature hazard's event, the explosion of reactor. By keeping finding the intermediate events, the basic events and their corresponding basic process control system elements are identified what all be formed the Fault Tree Analysis below by using 'or' gate and 'and' gate.[10]



After finding the basic control elements, their average possibilities of failure on demand (PFD) are researched by using Exida's research on *Failure Modes, Effects, and Diagnostic Analysis* and *Reliability Analytics Corporation's* web.[1][2] In this case, the top event's PFD is calculated by using 'Or gate' logic and 'And gate' logic which is 0.009. Now, since the later PFD is taken during 1 year test time interval, so the frequency or the demand rate can be found by using

$$\lambda = \frac{P * 2}{T} = 0.009 * \frac{2}{1} = 0.018 \text{ times/year}$$

In this case, the last part's hazard demand rate parameter should be W_1 what indicate the safety integrity level is 3.

However, by considering the low demand requirement, the safety integrity level is calculated by the formula

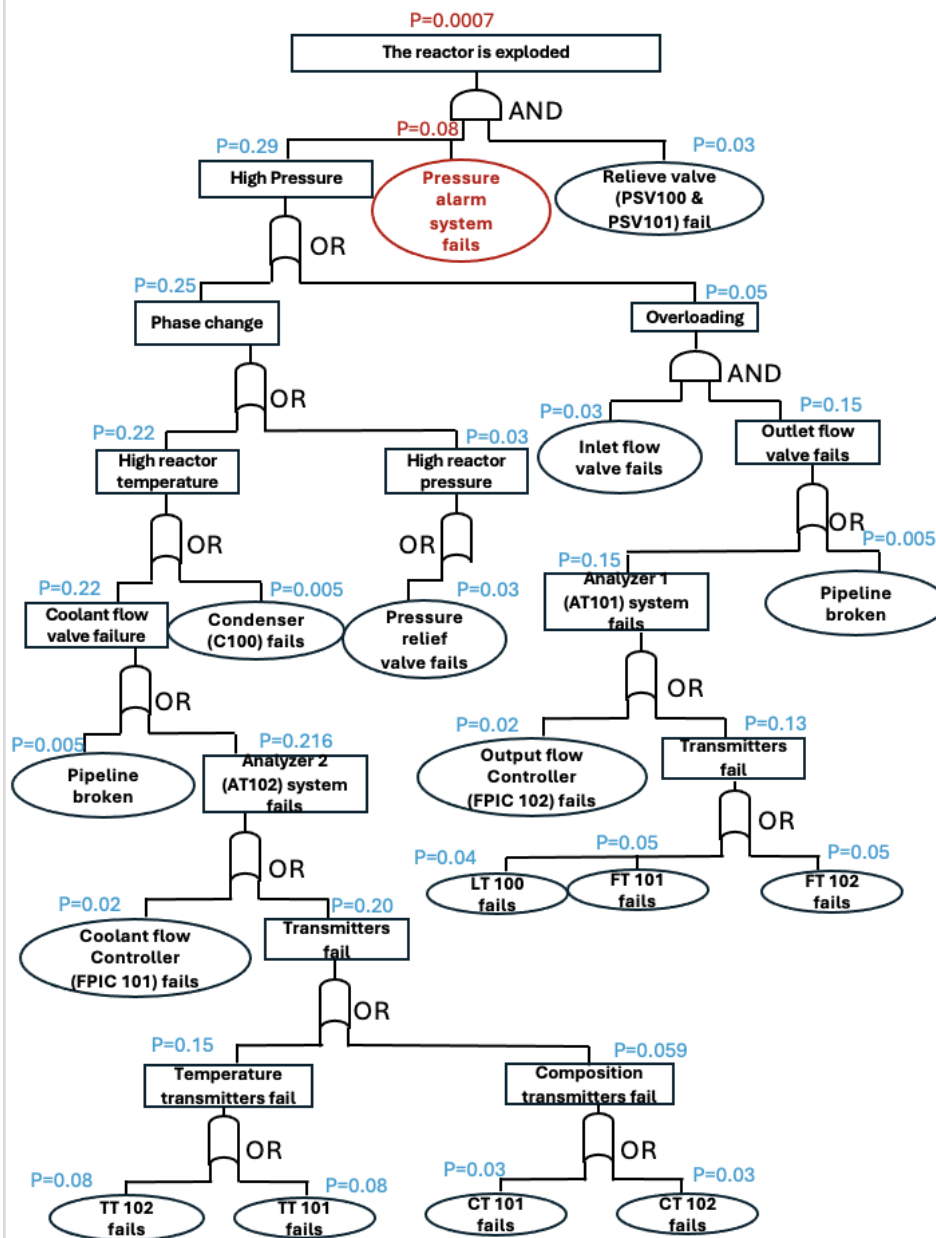
$$SIL = -\log_{10}(PFD_{avg}) \approx 2$$

Which is not meted the desired safety integrity level 3.[11]

4.4 Implementation of Safety Instrumented Function

To meet the target safety integrity level 3, the additional safety instrumented function needs to be applied to decrease the probability of failure on demand. To avoid high pressure, a more pressure alarm system can be added where the high pressure's failure probability can be efficiently decreased. In this case, the red part which is shown in the Figure 3.1 P&ID graph is the additional pressure safety instrumented function. By adding it, the reactor's explosion as the top event will have a new logic gate as 'And gate' what will highly decrease the PFD since possibility of both high pressure, pressure alarm system, and relieve valve failing at the same time will be less than only two event's PFD. Now, a new Fault Tree Analysis of protected system is shown in part 4.5.

4.5 Protected Fault Tree Analysis



By adding the additional implementation of safety function, the pressure alarm system, its average PFD is decreased from 0.009 to 0.0007 what indicates a new safety integrity level as level 3 by using the low demand formula

$$SIL = -\log_{10}(PFD_{avg}) = -\log_{10}(0.0007) \approx 3$$

and it meets the target to be safety integrity level 3 mentioned in part 4.3. [12]

4.6 Justification of the used failure rates

In the Part 4.3 and 4.5, the fault tree analysis used the failure rate of basic control elements to find the probability of failure on demand. However, their reliability and limitations should be justified to search the potential problems. For the reliability, the data all from Exia as the engineering safety solution company for over 20 years, and their solutions are following the IEC 61508 and IEC 61511 regulation. In this case, for example, the failure rate of temperature sensor is conducted by using Failure Modes, Effects, and Diagnostics Analysis (FEDA) to functionally and systemically. In the research, the test time interval is the lifetime (10 years) of the transmitters, in this long test time, it allows the engineers to detect more failures what make the final result to be more reliable. By considering test time interval to be 1 to 2 year, the number of detections will less than the 10 year's result which means high possibility of not detecting enough data of failure, and it makes the found failure rate to be less reliable. [12]

5 Sustainable Reflection

Your unit designs include one and two's benefits and potential negativities of socio-economy and social safety. During the unit designing of nitration of benzene reactor, a detailed result as a real process plant is formed by specifying the reaction kinetics, the reactor sizing, the control system, the safety system, and the safety integrity analysis. After the plant is constructed, its output chemical nitrobenzene is exclusively used around the world's chemical industry as the fundamental materials will bring the local society, the government, and the community more worker places, motivation of local economic growth, development of local supplier, generous tax, and technology transfer and innovations, so that the design plant can boost surround economic development, government benefits, and potential new innovations of sciences.

However, everything has two sides that a chemical plant certainly brings new economic opportunities but also some potential safety concerns to the local community. However, everything has two sides that a chemical plant certainly brings new economic opportunities but also some potential safety concerns to the local community due to the handling of hazardous materials and complex manufacturing processes. Accidental releases or spills of toxic or flammable chemicals can contaminate the environment and endanger human health. Fires, explosions, and toxic emissions are potential risks, along with transportation hazards related to the movement of materials. Process safety incidents, such as equipment failures or chemical reactions, can lead to catastrophic accidents if not properly managed. Additionally, environmental contamination from wastewater discharges and runoff can impact water bodies and soil quality. Effective emergency response preparedness is essential to minimize harm in the event of a chemical incident. Addressing these concerns requires regulatory compliance, risk assessment, hazard mitigation measures, and community engagement. By prioritizing safety and implementing appropriate safeguards, chemical plants can mitigate risks to local communities and operate responsibly within their surroundings.

6 Conclusion

The design of the unit encompasses critical aspects of reaction kinetics and reactor sizing, vital for chemical engineering industries. However, to operationalize these theoretical constructs, a bridge to reality is imperative, entailing considerations for control and safety systems. Identifying key disturbances such as variations in feed composition and ambient temperature is paramount, laying the groundwork for robust control strategies. Through feedback and feedforward mechanisms, sensitive variables like reactor temperature and liquid level are identified, alongside the nuanced discussion of control valve positions and controller types. The subsequent analysis delves into the intricate interplay between control loops, particularly focusing on reactor temperature and liquid level control. Notably, solutions for mitigating main disturbances are proposed, emphasizing the efficacy of feedforward control strategies.

Transitioning from theoretical frameworks to practical implementation, a comprehensive Pipe and Instrument Diagram (P&ID) is delineated, integrating essential components such as the Continuous-Stirred-Tank reactor, condenser, and output pump. Concurrently, safety instrumentation systems are meticulously incorporated, featuring emergency shutdown mechanisms, alarms, relief valves, and maintenance protocols. Further enhancing safety measures, the addition of a pressure Safety Instrumented Function is highlighted, reinforcing the P&ID's robustness. Subsequent safety integrity level analysis underscores the identification of hazards such as high reactor temperature and pressure, necessitating fault tree analysis to ascertain system reliability. Despite initial discrepancies, the integration of a pressure alarm system ensures compliance with target safety integrity levels, as corroborated by the protected fault tree analysis. Moreover, rigorous discussions on failure rates, reliability, and limitations provide a holistic understanding of potential operational challenges.

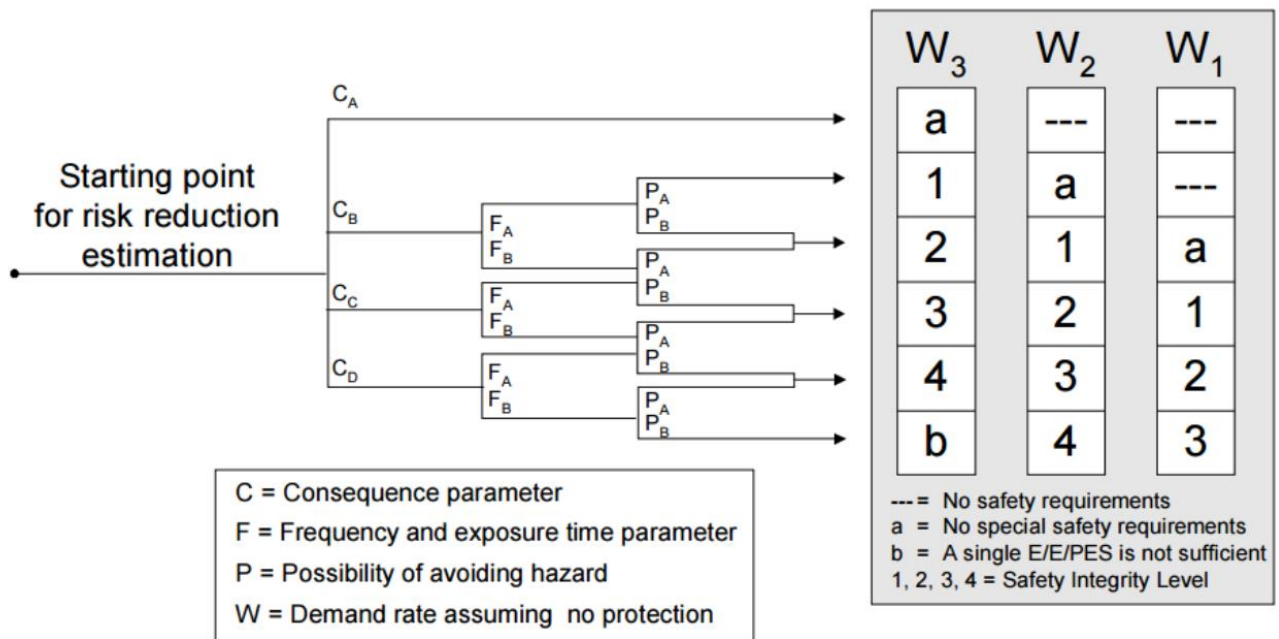
Beyond technical facets, the unit design's implications extend to socioeconomic and environmental domains. While it promises economic growth through job creation and local supply chains, it also underscores the imperative of regulatory compliance and community engagement to mitigate environmental and safety concerns. By aligning with regulatory standards, conducting comprehensive risk assessments, and fostering community dialogue, the unit design can effectively balance economic prosperity with environmental stewardship and safety assurance.

References

- [1] Raja, G. L., & Mohammad, S. (2023, November 22). Introductory Chapter: Introduction to Disturbance Rejection Control. Wwww.intechopen.com; IntechOpen. <https://www.intechopen.com/chapters/87555>
- [2] Ramnarine, E. (2023). Control Strategy: A Feedback-Feedforward “Controls Hub” for Risk, Knowledge, and Product Lifecycle Management. *Pharmaceutical Technology*, 47(10), 30–46. <https://www.pharmtech.com/view/control-strategy-a-feedback-feedforward-controls-hub-for-risk-knowledge-and-product-lifecycle-management#>
- [3] Types of Level Measurement Transmitters & How Do They Work? (n.d.). Wwww.transmittershop.com. Retrieved April 8, 2024, from <https://www.transmittershop.com/blog/level-measurement-transmitters-work/#:~:text=These%20transmitters%20are%20normally%20mounted>
- [4] Valve Failure Positions. (2014, December 17). Wwww.aiche.org. <https://www.aiche.org/ccps/resources/glossary/process-safety-glossary/valve-failure-positions#:~:text=In%20the%20event%20of%20instrument>
- [5] Dr Vipin Jain. (2012, February 24). Types of Controllers | Proportional Integral and Derivative Controllers | Electrical4U. Electrical4U. <https://www.electrical4u.com/types-of-controllers-proportional-integral-derivative-controllers/>
- [6] Stability of Control System - an overview | ScienceDirect Topics. (n.d.). Wwww.sciencedirect.com. <https://www.sciencedirect.com/topics/engineering/stability-of-control-system>
- [7] Ellis, G. (2002, January 1). Chapter 6 - The Luenberger Observer and Disturbances (G. Ellis, Ed.). ScienceDirect; Academic Press. <https://www.sciencedirect.com/science/article/abs/pii/B9780122374722500079>
- [8] Control systems. (2015). Hse.gov.uk. <https://www.hse.gov.uk/comah/sragtech/techmeascontsyst.htm>
- [9] Analysis, Synthesis, and Design of Chemical Processes Fourth Edition. (n.d.). <https://ptgmedia.pearsoncmg.com/images/9780132618120/samplepages/0132618125.pdf>
- [10] Gonzalez, Z. (n.d.). Fault Tree Analysis Guide with Example. SafetyCulture. <https://safetyculture.com/topics/fault-tree-analysis/>
- [11] Aschenbrenner, S. (n.d.). Failure Modes, Effects and Diagnostic Analysis. Retrieved April 7, 2024, from https://library.e.abb.com/public/fa9182bae8e54099b361441b12b0eaff/CE_TTX300_exida_FMEDA-EN.pdf?x-sign=kA9BDjicphH9ijPYSwFptx01hf3qhvA4VOMlZR0bHOTf2H1EYMoEkG8n5+ulhz5Q
- [12] Reliability Analytics Toolkit. (n.d.). Reliabilityanalyticstoolkit.appspot.com. Retrieved April 7, 2024, from <https://reliabilityanalyticstoolkit.appspot.com>

Appendix

Risk Graph



From L9_V5 SIL Selection